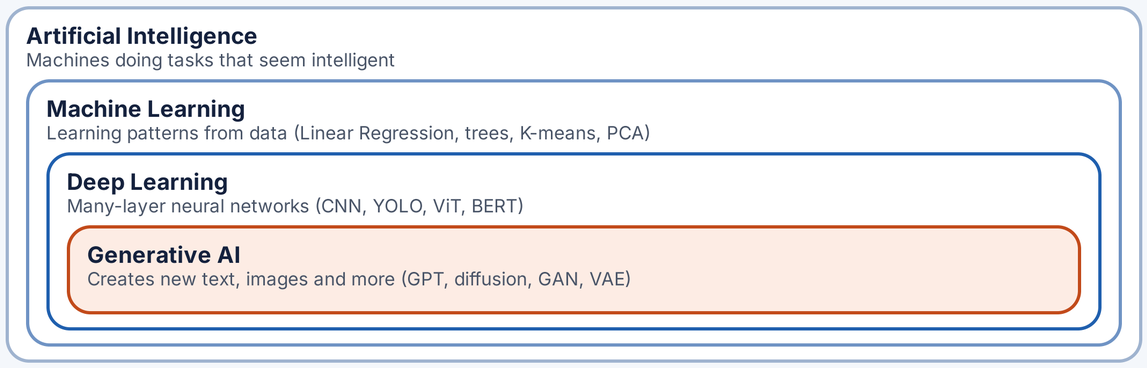
The Machine Learning Algorithm Map

15 algorithms sorted by what they work on (tabular data, images, text, new content) and how they learn (with labels, without labels, or from the data itself).

WHERE THEY SIT



How it learns: SUPERVISED uses labels · SELF-SUPERVISED makes its own labels · UNSUPERVISED finds patterns, no labels

Tags = also used in: TABULAR · VISION · LANGUAGE · GEN AI

Tabular Data
Rows and columns: sales, patients, customers

Linear Regression — SUPERVISED
How: fits the straight line that best matches the data, then uses it to predict a number.

Use: house prices, sales forecasts.

Decision Tree — SUPERVISED
How: asks a chain of yes/no questions about the features until it reaches an answer.

Use: loan approval, patient triage rules.

Random Forest — SUPERVISED
↳ builds on Decision Tree

How: grows many trees on random samples of the data and lets them vote.

Use: credit risk, customer churn.

Gradient Boosting — SUPERVISED
↳ builds on Decision Tree

How: adds small trees one at a time, each fixing the mistakes of the last.

Use: fraud detection, ad ranking (XGBoost).

Computer Vision
Photos, video and scans

CNN — SUPERVISED
How: slides small filters over an image to find edges, then shapes, then whole objects.

Use: medical imaging, face unlock, photo tagging.

YOLO — SUPERVISED
↳ builds on CNN

How: looks at the image once and predicts boxes and labels for every object in one pass.

Use: self-driving cars, store shelf scanning.

Vision Transformer — SUPERVISED
↳ builds on the Transformer (from language models)

How: cuts an image into patches and uses attention to see how the patches relate.

Use: image classification, scan analysis.

Natural Language
Emails, reviews, documents, chat

Naive Bayes — SUPERVISED
How: counts words in each class and uses Bayes' rule to pick the most likely one.

Use: spam filters, quick sentiment checks.

BERT — SELF-SUPERVISED
↳ Transformer encoder

How: learns by guessing hidden words, reading a sentence in both directions with attention.

Use: search ranking, text classification, Q&A.

GPT — SELF-SUPERVISED
↳ Transformer decoder

GEN AI

How: predicts the next word again and again, so it can write whole answers.

Use: chatbots, code assistants, summaries.

Generative AI
Creates new images, audio and designs

Diffusion Models — UNSUPERVISED
VISION

How: learns to remove noise, then starts from pure noise and cleans it up guided by a prompt.

Use: text-to-image art (Stable Diffusion), product mockups.

GAN — UNSUPERVISED
VISION

How: a generator makes fakes while a discriminator spots them, so both keep improving.

Use: synthetic faces, extra training images.

VAE — UNSUPERVISED
How: squeezes data into a small code and learns to rebuild it, then samples new codes to make new items.

Use: new molecule designs, anomaly detection.

Cross-domain tools that need no labels
These work on tabular data and on images alike, so they sit under all four columns.

K-means Clustering — UNSUPERVISED
TABULAR · VISION

How: places k centres, assigns each point to the nearest one, moves each centre to the middle of its group, and repeats until nothing changes.

Use: customer segments, compressing the colors in an image.

PCA — UNSUPERVISED
TABULAR · VISION

How: finds the few directions that hold most of the variation and squeezes many features down to those.

Use: plotting many-column data, removing noise, face recognition.

Which one should I reach for?

A table of numbers
Start with Random Forest or Gradient Boosting. Use Linear Regression when you need a simple, explainable number.

Images or video
CNN is the standard. YOLO for finding objects. ViT when you have a lot of data.

Text
Naive Bayes is a quick baseline. BERT to understand text, GPT to write it.

Make something new
Diffusion for images, GAN for sharp fakes, VAE for smooth variations, GPT for text.

No labels yet
K-means to find groups, PCA to simplify the data first.

Deepthi Pinninti | AIML Portfolio Artifact, Workshop 1.5

Simplified. Each algorithm sits where it is most often used. BERT and GPT are pre-trained self-supervised, then often fine-tuned with labels; ViT is supervised when trained on labeled images.